Existence of an aleph-fixed point via an omega-chain limit

The Open Logic Project contributors

openlogic:set-theory/card-arithmetic:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-card-arithmetic/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof's final equality chain is a single display; the superseded prose proof exists only in source comments (lines after the display), outside the claimed text.

Editorial note. The label alephfixed names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Union of an omega-chain of cardinals is a cardinal. Cardinals part, classing section: unioncardinalscardinal. Lines 702-703: kappa := union of the kappas-n is a cardinal.
- Aleph enumeration and transfinite recursion. Cardinals part: alpha |-> aleph_alpha as a class function; recursion theorem. The recursive definition of the sequence at lines 696-700 and the reindexing of the union in the closing display.

Source reference. Original reference unioncardinalscardinal: sources/olp-set-theory-card-arithmetic/pinned-original-source.tex, source lines 703–703. Named citation invoked at source line(s) 703: the author's marker \olref[cardinals][classing]{unioncardinalscardinal} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *There is an $\aleph$-fixed-point.*

Proof. Using recursion, define:

$$\begin{aligned} \kappa_0 &= 0 \\ \kappa_{n+1} &= \aleph_{\kappa_n} \\ \kappa &= \bigcup_{n<\omega} \kappa_n \end{aligned}$$

Now κ is a cardinal by unioncardinalscardinal . But now:

$$\kappa = \bigcup_{n<\omega} \kappa_{n+1} = \bigcup_{n<\omega} \aleph_{\kappa_n} = \bigcup_{\alpha<\kappa} \aleph_\alpha = \aleph_\kappa$$

□